IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS

EEE PART III/IV: MEng, BEng and ACGI

ELECTRICAL ENERGY SYSTEMS

Wednesday, 10 December 2025, 10:00 AM

Time allowed: 3:00 hours

There are SIX questions on this paper.

Answer any 2 out of 3 questions from Section A and answer any 2 out of 3 questions from Section B.

All questions carry equal marks.

Formula sheet will be provided.

Any special instructions for invigilators and information for candidates are on page 1.

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Second Marker(s) : B. Chaudhuri, F. Teng

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Section A: Answer any two questions out of Q1 to Q3

1. **Question on power system design and frequency regulation**

a) Assume that there are n identical generators, each with a capacity of 50 kW and availability of 0.98.

i) Explain the benefits of developing large-scale interconnected power systems for electricity supply.

[2 marks]

ii) The system design standard requires that the probability of meeting P kW peak demand is equal or greater than X. As a system design engineer, formulate a general optimization problem to determine the minimum number of the generators n_{min} that meet this standard (P is a parameter in the optimization).

[4 marks]

iii) Assume that the coincidence factor of an infinite number of households is 0.333, the peak demand of each individual household is 4 kW, and the reliability criterion requires that the probability of meeting peak demand is equal to or greater than 99%. Based on the formulated optimization problem, find n_{min} to supply 90 households.

[4 marks]

b) Battery storage systems can provide frequency response much faster than thermal generators, therefore a new service called Dynamic Containment (DC) is designed in the UK. DC can effectively reduce the need for the regular frequency response (R) to maintain the frequency nadir as required, following the rule below:

$$\left(\frac{H}{f_0} - \mathrm{DC} \times \frac{T_{\mathrm{DC}}}{4 \times \Delta f_{\max}}\right) \times R \geq \frac{(P_L - \mathrm{DC})^2 \times T_d}{4 \times \Delta f_{\max}} \tag{1.1}$$

Nomenclature:

H: System level of inertia (MW-s)
R: Amount of frequency response scheduled (MW)
DC: Amount of DCscheduled (MW)
P_{L}: Value of generation outage we want to secure against (MW)
f_0: Nominal frequency of the power grid (Hz)
$\Delta f_{\max}$: Maximum admissible frequency deviation (Hz)
$\mathrm{T_d}$: Delivery time of frequency response (s)
T_{DC}: Delivery time of DC (s)

Assumptions:

- The amount of frequency response: $R = 2500$ MWs,
- The cost of frequency response: £5/MW,
- The largest generation unit in the UK: 1.8 GW,
- Other parameters: $f_0 = 50$ Hz, $\Delta f_{\max} = 1.0$ Hz, $T_d = 10$ s, $T_{\mathrm{DC}} = 0.5$ s.

i) Please list three key index to describe the post-fault system frequency and discuss the potential reasons that the system may encounters large (in magnitude) deviations in those three index?

[3 marks]

ii) Based on (1.1), calculate the minimum required inertia in the system to maintain the frequency nadir constraint. Do the same calculation when DC =200 MW is added to replace 200 MW frequency response R. Please comment the differences in the inertia response in those two cases

[3 marks]

iii) Based on the case with minimum required inertia and 0 MW DC in (ii), assuming there is a bid of 150MW of DC, what is the maximum price (£/MW) the system operator should pay for it to achieve the minimum cost?

[4 marks]

2. **Question on per unit system**

a) What are the advantages to apply per-unit system in the analysis of power systems?

[3 marks]

b) Consider a three-phase high-voltage transmission circuit in Figure 2.1. A 60 MW, 0.8 lagging power-factor load is supplied from a 33 kV substation whose receiving bus is regulated at 30 kV.

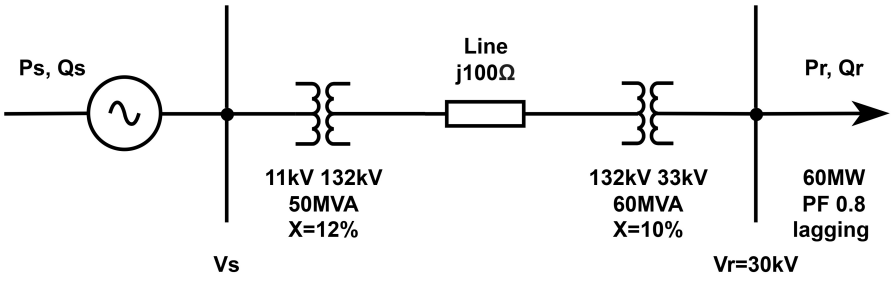


Figure 2.1: High-voltage transmission circuit with specified parameters.

i) Calculate the per-unit values of the quantities (three impedances) in the figure with properly selected bases for convenience, and sketch the equivalent circuit.

[5 marks]

ii) Calculate the magnitude and phase angle of the voltage V_s at the sending end.

[5 marks]

iii) At the receiving end, when
(1) $P_r = 0$ and Q_r are set to 1 p.u. and 0.5 p.u. respectively,
(2) $Q_r = 0$ and P_r are set to 1 p.u. and 0.5 p.u. respectively,
calculate the magnitude and phase angle of the voltage V_s in these four scenarios again and state your findings.

[7 marks]

3. **Question on power flow analysis and Gauss-Seidel method**

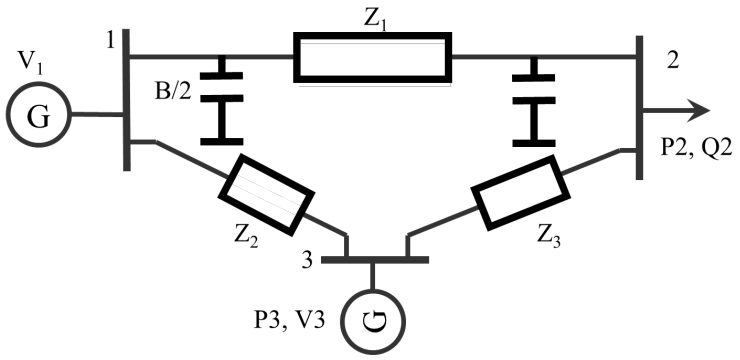


Figure 3.1: A three-bus power network

A three-bus power network is presented in Figure 3.1.

a) Express the bus admittance matrix for the system in terms of Z_1, Z_2, Z_3 and B.
[5 marks]

b) Assuming the per unit values $|V_1| = 1, |V_3| = 1.05, P_3 = 1.03, P_2 = 0.96, Q_2 = 1, Z_1 = \mathrm{j}0.18, Z_2 = \mathrm{j}0.20, Z_3 = \mathrm{j}0.18, B = 0$, perform two iterations of the Gauss-Seidel load flow method.
[12 marks]

c) What factors could lead to a high voltage magnitude at Bus 3, and what mitigation measures would be taken in this case?
[3 marks]

Section B: Answer any two questions out of Q4 to Q6

4. **Question on balanced fault analysis**

a) Discuss how the system short-circuit level would change under the following scenarios:

i) When an AC interconnection is established between two independent power systems.

[1 mark]

ii) When some gas-fired generation plants are replaced by equivalent inverter based resources.

[1 marks]

b) Justify the use of the sub-transient reactance of a synchronous generator in short-circuit analysis.

[4 marks]

c) Two 275 kV AC systems are to be linked through a pair of identical parallel transmission lines, each with a reactance of 0.1 p.u. on a 1000 MVA base. Before interconnection, the short-circuit levels at the respective connection points are 5000 MVA and 7500 MVA. Using a common system base of 1000 MVA, neglect resistance and assume prefault voltages of 1.0 p.u.

i) Determine the short-circuit levels at the interconnection buses after the two systems are connected.

[7 marks]

ii) Determine the three-phase fault current (in kA) for a zero-impedance fault at the midpoint of one of the lines.

[7 marks]

You may use the following formula for delta (Δ) to star (Y) conversion:

$$Z_A = \frac{Z_{AB}Z_{CA}}{Z_{AB}+Z_{BC}+Z_{CA}}, \quad Z_B = \frac{Z_{AB}Z_{BC}}{Z_{AB}+Z_{BC}+Z_{CA}}, \quad Z_C = \frac{Z_{BC}Z_{CA}}{Z_{AB}+Z_{BC}+Z_{CA}}.$$

where the impedances of the three phases of a delta connection are Z_{AB}, Z_{BC}, Z_{CA}, and the impedances of the equivalent star connection are Z_A, Z_B, Z_C.

5. **Question on unbalanced fault analysis**

a) The primary side of a delta-connected transformer draws the following line currents from a generator:

$$I_B = 0.70\angle 50^\circ \text{ p.u.}, \qquad I_C = 0.82\angle 34^\circ \text{ p.u.}.$$

Calculate the magnitude (in p.u.) and phase angle (in degrees) of the positive-sequence, negative-sequence and zero-sequence components of the line current flowing from the generator to the transformer.

[6 marks]

b) For the single-line diagram in Figure 5.1, the sequence reactances of the components in p.u. on a 500 MVA base are given in Table 5.1.

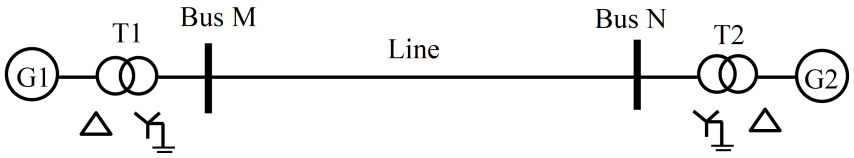


Figure 5.1: Single-line diagram of the three-phase power system.

Table 5.1: Positive, negative and zero sequence reactances on 500 MVA base

Component	X_1 (p.u.)	X_2 (p.u.)	X_0 (p.u.)
Generators G1, G2 (13 kV)	0.12	0.12	0.02
Transformers T1, T2 (13/133 kV)	0.15	0.15	0.15
Transmission line (133 kV)	0.07	0.07	0.18

A line-to-ground fault with zero fault impedance occurs on phase A at bus N. Neglect resistance and assume pre-fault generator voltages are 1.0 p.u. and pre-fault currents are zero.

i) Compute the total current (in kA) flowing into the fault at bus N.

[5 marks]

ii) Determine the phase-B voltage magnitude (in kV) at bus N during the fault.

[5 marks]

iii) Calculate the current (magnitude and angle) flowing in phase A of the line.

[4 marks]

6. **Question on stability analysis**

a) The maximum power that can be transmitted over an uncompensated transmission line decreases as the transmission distance increases. Sketch a representative line loadability curve for transmission distances from 0 to 600 miles, and discuss the principal factors that limit the permissible power transfer over different distance ranges (short, medium, and long).

[4 marks]

b) Starting from the sinusoidal power-angle characteristic and the swing equation:

$$P_e = \frac{EV}{X}\sin\delta, \quad \frac{2H}{\omega_s}\frac{d^2\delta}{dt^2} = P_m - P_e,$$

derive an expression for the natural frequency of oscillation of a round-rotor synchronous generator in terms of the synchronising coefficient ($\frac{dP_e}{d\delta}$) and the inertia constant H.

[5 marks]

c) A 500 MVA, 50 Hz round-rotor synchronous generator has synchronous reactance, $X_s = 0.2$ p.u. and inertia constant, $H = 4$ MW·sec/MVA. The generator is connected to a large power system through a transformer and overhead line having a combined reactance of 0.3 p.u. on a 500 MVA base. The voltage magnitude at both the generator terminals and the system connection point is 1.0 p.u. The generator supplies 450 MW to the system. Resistances are neglected.

i) Determine the internal voltage (in p.u.) of the generator behind its synchronous reactance.

[3 marks]

ii) Compute the natural frequency of oscillation (in Hz) of the generator under these conditions.

[4 marks]

iii) Using equal area criterion, determine the critical clearing angle (in degrees) for a three-phase fault at the generator terminals assuming the generator output is zero during the fault.

[4 marks]

ELEC60012 – Electrical Energy Systems

Formula sheet

Part A

1. Coincidence coefficient for *n* households

$$j_n = j_\infty + \frac{1 - j_\infty}{\sqrt{n}}$$
$$P_n = j_n \times n \times P_1$$

2. Gauss-Seidel power flow update law

$$PQ\ bus: V_i^k = \frac{1}{Y_{ii}}\left(\frac{S_i^*}{(V_i^{k-1})^*} - \sum_{j=1}^{i-1} Y_{ij}\, V_j^k - \sum_{j=i+1}^{Nbus} Y_{ij}\, V_j^{(k-1)}\right)$$

Part B

1. Short circuit level in MVA

$$SCL\ (\text{in MVA}) = \frac{SCL_{base}\ (\text{in MVA})}{X_{eq}\ (\text{p.u.})}$$

2. Transformation matrix for sequence domain (a-phase) to phase domain and vice versa

$$\begin{bmatrix} V_a \\ V_b \\ V_c \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & \alpha^2 & \alpha \\ 1 & \alpha & \alpha^2 \end{bmatrix} \begin{bmatrix} V_{a0} \\ V_{a1} \\ V_{a2} \end{bmatrix}$$

$$\begin{bmatrix} V_{a0} \\ V_{a1} \\ V_{a2} \end{bmatrix} = \frac{1}{3}\begin{bmatrix} 1 & 1 & 1 \\ 1 & \alpha & \alpha^2 \\ 1 & \alpha^2 & \alpha \end{bmatrix} \begin{bmatrix} V_a \\ V_b \\ V_c \end{bmatrix}$$

The same transformation matrices are valid for current as well

3. Swing equation

$$\left(\frac{2H}{\omega_s}\right)\frac{d^2\delta}{dt^2} = P_m - P_e\ \text{p.u.}$$